

The Beneficial Effects of Raffinee in Permanent Occulted Stroke Mice

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ABSTRACT Ischemic stroke is a major cause of disability and mortality globally. Although thrombolytic therapy is routinely adopted in cases of ischemic stroke, various alternative natural neuroprotectants are also used as effective adjuvant therapies to recover neurofunction following ischemic stroke. Raffinee, a natural fermented product with strong antioxidant and neuroprotective activities, has antiatherogenic effects in animals and has exhibited neuroprotective effects in a clinical trial by recovering motor and sensory function following spinal cord lesion. This study reveals the advantageous effects of Raffinee on PC12 cells by decreasing hypoxia-induced apoptosis in mice with permanent middle cerebral artery occlusion (pMCAO) by increasing the levels of neurotrophic factors such as S100 β , reducing serum inflammatory factors such as matrix metalloproteinases (MMP)-9/MMP-2 ratio, tumor necrosis factor- α , and interleukin (IL)-6 level, and increasing IL-10 levels. Significantly reduced brain infarct volume along with a favorable survival ratio was observed for pMCAO mice that received Raffinee, suggesting a neuroprotective potential of Raffinee in cases of acute ischemic stroke by suppressing apoptosis.

KEYWORDS: • inflammatory cytokines • ischemic stroke • permanent middle cerebral artery occlusion • Raffinee

INTRODUCTION

EVIDENCE INDICATES THAT OVER five to six million people die from stroke globally every year.¹ The burden of stroke is high, and the numbers of stroke patients are increasing.² Ischemic stroke represents nearly 87% of stroke cases³; it is a dominant reason for disability and mortality worldwide.⁴ Various pathological conditions, including excitotoxicity, excessive production of reactive oxygen species, mitochondrial dysfunction, inflammation, neuronal apoptosis, and blood–brain barrier permeability, have been linked to cerebral injury during ischemic stroke.^{5,6} Many inflammation-related proteins such as matrix metalloproteinases (MMPs), C-reactive protein, tumor necrosis factor (TNF)- α , interleukin (IL)-6 and IL-10, and neurotrophic factors, such as S100 β and neuron-specific enolase (NSE),^{7–14} are also associated with the diagnosis and prognosis of ischemic stroke, suggesting the involvement of complex pathological processes therein.

Thrombolytic therapy and neuroprotective therapy are two major therapeutic strategies for ischemic stroke.^{15,16} The administration of tissue-plasminogen activator is the

main therapeutic intervention for ischemic stroke. However, less than 2% of ischemic stroke patients have the opportunity to receive thrombolytic therapy.¹⁷ Since thrombolytic therapy provides no substantial improvement after dissolving of the clot and blood flow reperfusion in either animals or humans with ischemic stroke, treatments with neuroprotectants are required to recover and protect motor and cognitive abilities in the long run.^{18,19} Therefore, alternative therapeutic approaches with significant neuroprotective effects for patients with ischemic stroke are urgently needed. For decades, substantial evidence has indicated that complementary and alternative medicine is helpful in the prevention of, and recovery from, ischemic stroke.^{20–22}

A natural fermented product of green tea, rice germ, soybean, and coffee bean, called Raffinee, has been reported to significantly improve and restore the motor and sensory function in patients with spinal cord injuries (SCI), based on an open-labeled clinical trial,^{23,24} suggesting the potential neuroprotective and functional recovery effects of Raffinee. Hence, this investigation examines the influence of Raffinee on the brain injury that occurs in ischemic stroke mice.

MATERIALS AND METHODS

Cells and hypoxia treatment

PC12 cells, a widely used model for neural studies,²⁵ were obtained from ATCC (PC-12; ATCC® CRL-1721™) and

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cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum. For hypoxia treatment, PC12 cells were washed and replaced with glucose-free DMEM. A modular incubation chamber (Billups-Rothenberg, USA) flushed with 95% nitrogen/5% CO₂ was then used to culture PC12 cells for 4 h at 37°C. For normal control (normal), PC12 cells were maintained in a normoxic chamber for 4 h after washing with glucose-containing DMEM.

Raffinee

Raffinee was obtained from Ziel Enterprise Co., Ltd., Kaohsiung, Taiwan. Production of Raffinee was described in a previous publication.²⁶ Briefly, fresh Indonesian arabica coffee beans were ground into powder, and the extract of the powder was percolated with water (pH 5.5). The coffee bean extract (20%), rice bran (3.5%), and soybean (0.1%) were then pooled together and added with water to 100%. Subsequently, the pH value of the mixed solution was adjusted to 9 and fermented with *Bacillus subtilis* Natto for 18 h. The main components of Raffinee are GMT (Guardox™), which contains polyphenols (flavonols, genistein, and chlorogenic acid), proteins, minerals, and carbohydrates. Each milliliter Raffinee contains 750 mg GMT.

MTT assay

Cell survival was detected using 3-(4,5-dimethyl-thiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) assay. Raffinee doses adopted in cell experiments were determined according to the dose used in animal experiments in this study. A total of 5×10^5 PC12 cells were preincubated in each well of a 24-well plate. Cells were under hypoxia under various concentrations of Raffinee (3.75 and 11.25 mg/mL) for 4 h and incubated in 0.5 mg/mL MTT at 37°C. Two hundred microliters of dimethyl sulfoxide was used to dissolve the crystal, and the absorbance was detected at 570 nm with an ELISA reader (SpectraMax M5, USA).

Protein extraction and immunoblotting

Cells were harvested by centrifugation and subsequently lysed in 600 µL of PRO-PREPTM solution (iNtRON Biotech, Korea). The protein supernatant was collected by centrifugation, and its concentration was detected using a spectrophotometer (Hitachi U3000, Japan) at 595 nm with Bovine Serum Albumin (BSA) as standard. Next, extracted proteins were spread into a 12% sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) gel and transferred to an NC paper (Bio-Rad, USA). After blocking with 1% gelatin in 1× phosphate-buffered saline (PBS), antibodies against Bcl-2, Bax, Cytokine c, Apaf-1 (Santa Cruz, USA), or β-actin (Chemicon, USA) were incubated with the NC papers for 2 h with gentle vibration. The papers were then subsequently reacted with horseradish peroxidase (HRP)-conjugated secondary antibody for another hour. The Immobilon Western Chemiluminescent HRP Substrate (Millipore, USA) and chemiluminescence imaging an-

alyzer (GE ImageQuant TL 8.1, USA) were used to measure the complexes of antigen and antibody.

Animals

Ninety-six male, 6-week-old C57BL/6J mice obtained from the National Laboratory Animal Center, Taiwan, were housed at Chung Shan Medical University, Taichung, Taiwan, under the supervision of the Institutional Animal Care and Use Committee (IACUC approval no.: 1656). All animals were kept at 23°C ± 2°C under a 12-h light-dark cycle and had free access to water and diet (Lab Diet 5001, USA) for 2 weeks.

Permanent middle cerebral artery occlusion

Since it is known to be an extremely useful model for studying ischemic stroke, we used permanent middle cerebral artery occlusion (pMCAO) for this study.²⁷ To produce focal cerebral ischemia, pMCAO in mice was performed as published elsewhere.^{28–30} First, mice were anesthetized with Thiopental (40 mg/kg) and placed on a heating pad at around 37°C ± 0.5°C during surgery. A wound was incised from ear to eye, and an orifice was made over the distal part of the middle cerebral artery. The middle cerebral artery was then electrocoagulated, and the wound was sewed using nylon-suture (#4.0). The sham-operations of the control mice were identical to those in the experimental groups except that the middle cerebral artery was not electrocoagulated. Following the operation, all animals were treated with isotonic saline. For post-surgical analgesia, all animals were subcutaneously injected with Buprenorphine (Buprenex™) (2 mg/kg).

Animal treatments

Raffinee is a natural fermented product with superoxide dismutase (SOD)-like antioxidant properties that was produced jointly by Ziel Enterprise and Toyo Hakko Company.^{23,24} Each milliliter of Raffinee consists of 750 mg of GMT (Guardox) and 33.75 mg of polyphenols.²⁴ The Raffinee dose used in animals of this study is according to the human dose²⁴ that has been converted based on the U.S. FDA Guidance for Industry.³¹ The date for validating the effects of Raffinee on mice after pMCAO is according to previous studies.^{30,32,33} At 8 weeks old, the mice were separated into the following eight groups ($n = 12$ per group); control mice (sham-operation) treated with PBS for 7 days (group A), control mice (sham-operated) treated with Raffinee for 7 days (group B), pMCAO mice treated with PBS for 1 day (group C), pMCAO mice treated with Raffinee for 1 day (group D), pMCAO mice treated with PBS for 3 days (group E), pMCAO mice treated with Raffinee for 3 days (group F), pMCAO mice treated with PBS for 7 days (group G), and pMCAO mice treated with Raffinee for 7 days (group H). Groups A and B were orally gavaged with 0.15 mL PBS and 0.15 mL Raffinee for 7 days, respectively. Groups C, E, and G were orally gavaged with 0.15 mL PBS immediately after pMCAO for 1, 3, and 7 days, respectively. Groups D, F, and H were

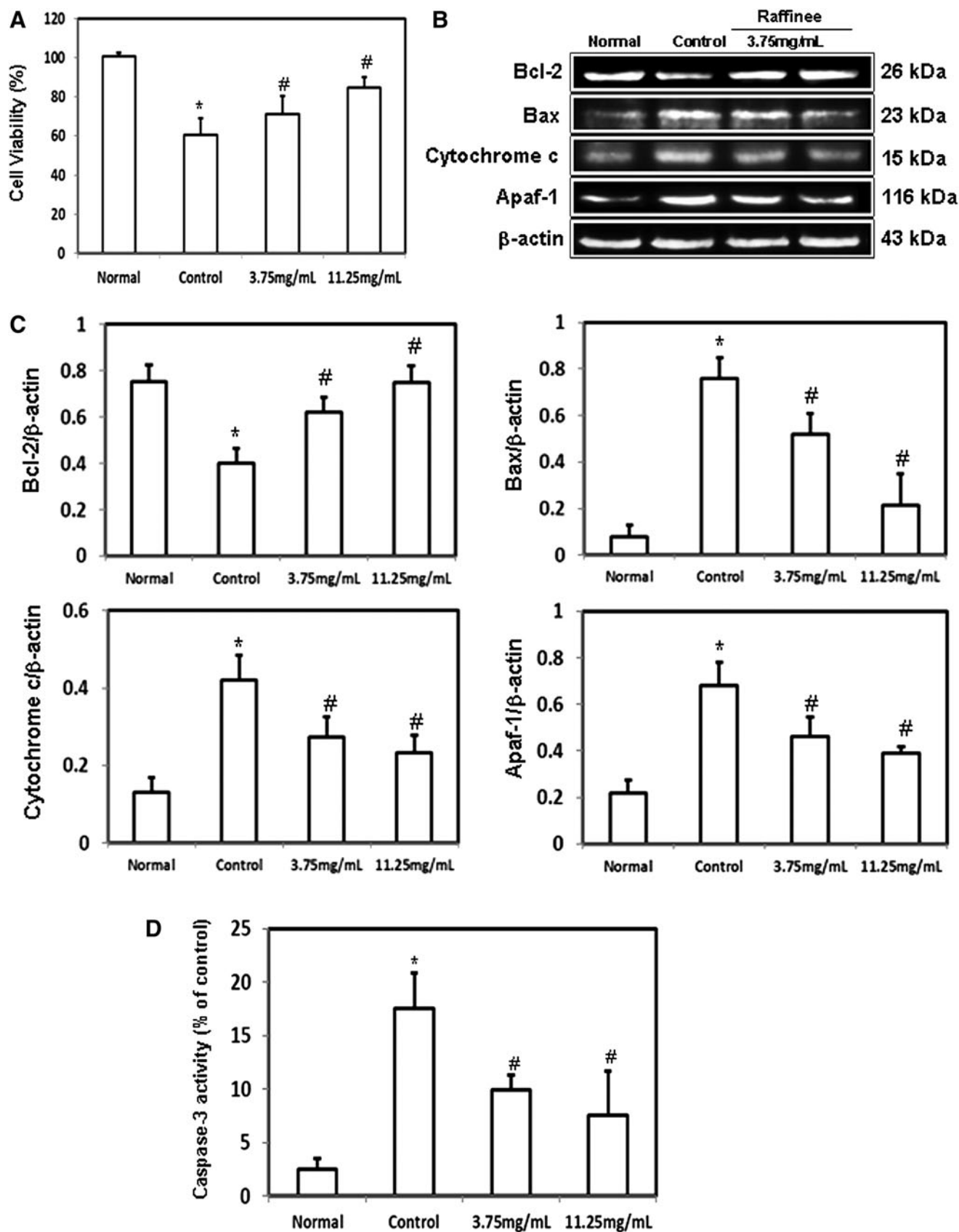


FIG. 1. Effects of Raffinee on PC12 cells after hypoxia treatment. **(A)** Viability of PC12 cells in the presence of Raffinee. **(B)** Expression of Bcl-2, Bax, cytochrome c, and Apaf-1 proteins in PC12 cells. **(C)** Relative levels of Bcl-2, Bax, cytochrome c, and Apaf-1 are based on β -actin expression. **(D)** Caspase 3 activity in PC12 cells. Data are presented as mean $\pm$ SEM (mean of standard deviation). Similar results were obtained in three repeated experiments. The symbols, * and #, indicate significant difference, $P \leq .05$, relative to the cells from the normal group and control group, respectively. SEM.

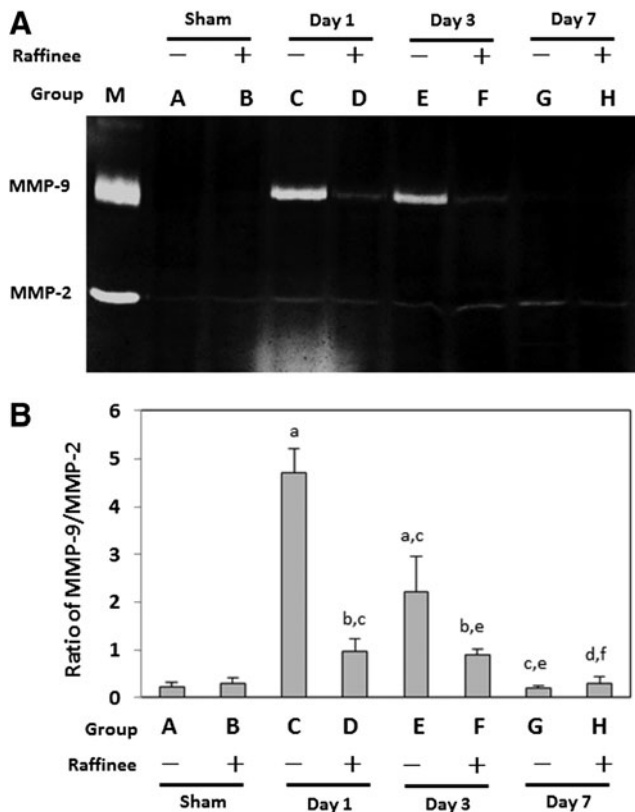


FIG. 2. Effects of treatment with Raffinee on the activity of MMP in mice after pMCAO. (A) Activity of serum MMP-9 and MMP-2 and (B) MMP-9/MMP-2 ratio in mice following pMCAO for 1, 3, or 7 days. Data are presented as mean $\pm$ SEM, $N=12$. The subscripts a, b, c, d, e, and f indicate significant difference compared with groups A, B, C, D, E, and F, respectively. MMP, matrix metalloproteinase; pMCAO, permanent middle cerebral artery occlusion.

orally gavaged with 0.15 mL Raffinee immediately after pMCAO treatment for 1, 3, and 7 days, respectively.

Infarct volume measurement

The mice were sacrificed by asphyxiation using CO₂, and their heart blood was rapidly harvested and their brains were quickly and completely removed. The brains of mice were continuously sectioned into six coronal sections (1 mm thickness). Infarct volumes were stained with 2,3,5-triphenyltetrazolium chloride.³⁰ Briefly, the brain slices were soaked in 4% paraformaldehyde for 1 day. The slices were then photographed using a computer-controlled digital camera (Olympus, DP70), and cerebral infarct area and volume were evaluated using a Fujifilm Multi Gauge V3.0 and Image-Pro Plus (IPP6.0). The area of brain infraction was determined according to a previous publication.³⁴

Gelatin zymography

MMP-2 and MMP-9 activities were detected according to a previous publication.³⁵ One microliter of mouse serum was diluted 10 times and separated in an 8% SDS-PAGE gel containing 1 mg/mL gelatin. After washing with 2.5% Tri-

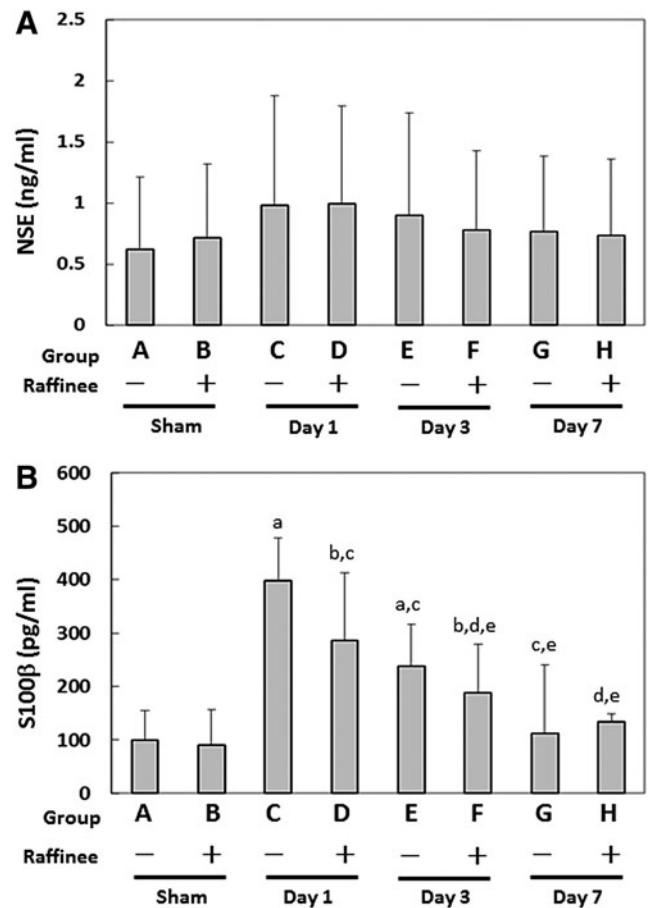


FIG. 3. Effects of treatment with Raffinee on expression of neurotrophic factors in mice following pMCAO. Serum level of (A) NSE and (B) S100 calcium-binding protein beta (S100β) in mice after pMCAO for 1, 3, or 7 days. Data are presented as mean $\pm$ SEM, $N=12$. Subscripts a, b, c, d, and e indicate significant difference compared with groups A, B, C, D, and E, respectively. NSE, neuron-specific enolase.

ton X-100 for 30 min, the gel was immersed in reaction solution at 37°C for overnight.³⁵ Gelatinolytic area was detected by staining with 0.5% Coomassie Brilliant Blue and calculated by densitometry (Appraise, USA).

ELISA

The caspase-3 activity (BD Pharmingen, CA, USA) in PC-12 cells and serum levels of S100β (Lifespan Biosciences, USA), NSE, TNF-α, IL-6, and IL-10 (Abcam, United Kingdom) were detected with ELISA kits.

Statistics

All value presentation and statistics were generated using GraphPad Prism 5 software (GraphPad Software). The comparisons among groups were made by one-way analysis of variance with Bonferroni *post hoc* tests followed by Tukey multiple comparison tests. All values are shown as mean $\pm$ SEM (mean of standard deviation). A P value smaller than .05 was considered statistically significant.

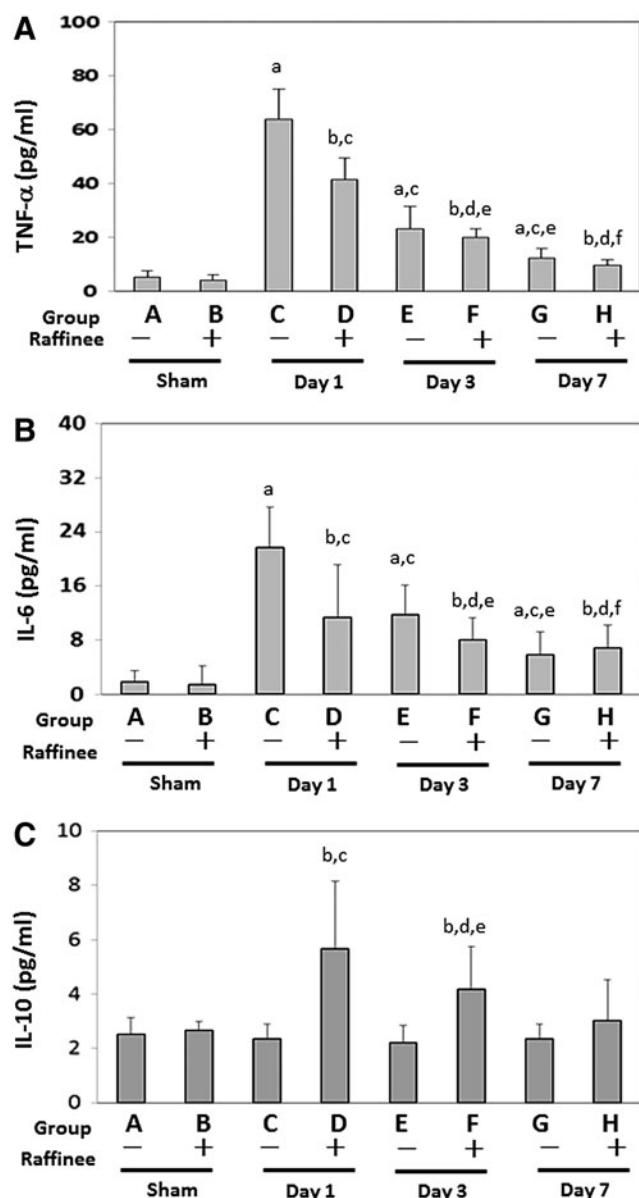


FIG. 4. Effects of treatment with Raffinee on expression of inflammatory associated cytokines in mice following pMCAO. Serum level of (A) TNF- α , (B) IL-6, and (C) IL-10 in mice after pMCAO for 1, 3, or 7 days. Data are presented as mean $\pm$ SEM, $N=12$. Subscripts a, b, c, d, e, and f indicate significant difference compared with groups A, B, C, D, E, and F, respectively. IL, interleukin; TNF- α , tumor necrosis factor- α .

RESULTS

Effects of Raffinee on PC12 cells under hypoxia

To assess the effects of Raffinee on PC12 cells under hypoxia, cell survival, apoptotic protein expression, and caspase-3 activity were measured. Significantly decreased viability of PC12 cells was observed under hypoxia compared with the normal group, whereas significantly higher PC12 cell viability was detected in the presence of Raffinee (5 and 15 μ L/mL) compared with the control group (Fig. 1A). Significantly lower Bcl-2 and higher levels of apoptotic mole-

cules, including Bax, cytochrome c, and Apaf-1, were detected under hypoxia compared with the normal group. Conversely, significantly increased Bcl-2 and decreased apoptotic molecules were detected in the presence of Raffinee (5 and 15 μ L/mL) compared with the control group (Fig. 1B, C). Moreover, significantly increased caspase-3 activity was detected in PC12 cells under hypoxia compared with the normal group, whereas significantly decreased caspase-3 activity in PC12 cells was detected in the presence of Raffinee (5 and 15 μ L/mL) compared with the control group (Fig. 1D).

Effects of MMPs and neurotrophic factors in pMCAO mice

To assess the effects of Raffinee on MMPs and neurotrophic factors in pMCAO mice, MMP-9 and MMP-2 activities and serum levels of NSE and S100 β of mice were detected. No statistical difference in the MMP-9/MMP-2 activity was detected between control mice (sham-operation) that had been treated with PBS (group A) and those treated with Raffinee (group B) for 7 days, or pMCAO mice treated with PBS (group G) and those treated with Raffinee (group H) for 7 days (Fig. 2A, B). A significantly increased MMP-9/MMP-2 activity was revealed in pMCAO mice that had been treated with PBS for 1 (group C) or 3 (group E) days (Fig. 2A, B), whereas a significantly lower MMP-9/MMP-2 activity ratio was observed in pMCAO mice administrated with Raffinee for 1 (group D) or 3 (group F) days. No significant difference in serum NSE and S100 β levels was detected between control mice (sham-operation) that had been treated with PBS (group A) and those treated with Raffinee (group B) for 7 days. No variation in serum NSE level was observed among all groups (Fig. 3A). Notably, significantly increased serum S100 β was observed in pMCAO mice receiving PBS for 1 (group C) or 3 (group E) days, whereas significantly reduced serum S100 β were detected in pMCAO mice receiving Raffinee for 1 (group D) or 3 (group F) days, respectively (Fig. 3B).

Effects of Raffinee on inflammation-related cytokines in pMCAO mice

To test the influence of Raffinee on inflammation-associated cytokines in pMCAO mice, serum TNF- α , IL-6, and IL-10 were detected. Significantly higher levels of serum TNF- α and IL-6 were detected in mice from groups C, E, and G than in those from group A (Fig. 4A, B). Significantly lower levels of serum TNF- α and IL-6 were detected in mice from groups D and F than in those from groups C and E, respectively (Fig. 4A, B). Significantly higher levels of serum IL-10, an anti-inflammatory cytokine, were detected in mice from groups D and F than in those from groups C and E, respectively (Fig. 4C).

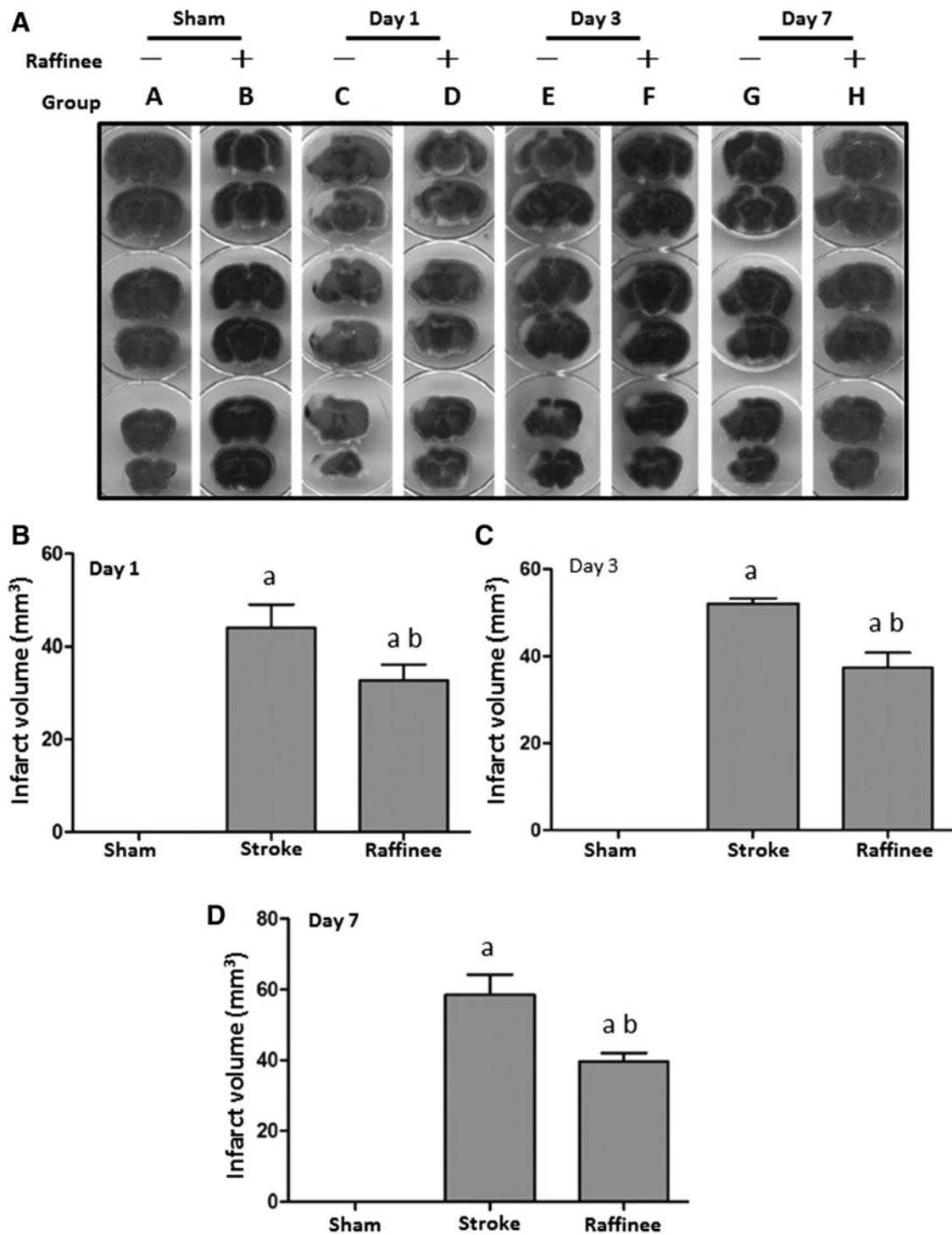


FIG. 5. Effects of treatment with Raffinee on infarct volume of mice following pMCAO. (A) Representative TTC-stained sections. (B–D) Infarct volume of mice in each group after pMCAO. Data are presented as mean infarct volume $\pm$ SEM, $N=12$. Subscripts a and b indicate significant difference compared with the sham and stroke groups, respectively. TTC indicates 2,3,5-triphenyltetrazolium chloride.

Effects of Raffinee on brain infarct volume in, and survival ratio of pMCAO mice

To study the effects of Raffinee on brain injury in pMCAO mice, infarct volume was measured. No difference was observed between brain coronal sections from the mice

from groups A and B. The infarct areas in the brain sections of mice from groups D, F, and H were smaller than those of mice from groups C, E, and G, respectively (Fig. 5A). The brain infarct volumes in the pMCAO mice that had been treated with Raffinee for 1, 3, or 7 days (groups D, F, and H) were lower than those in the controls that had been treated

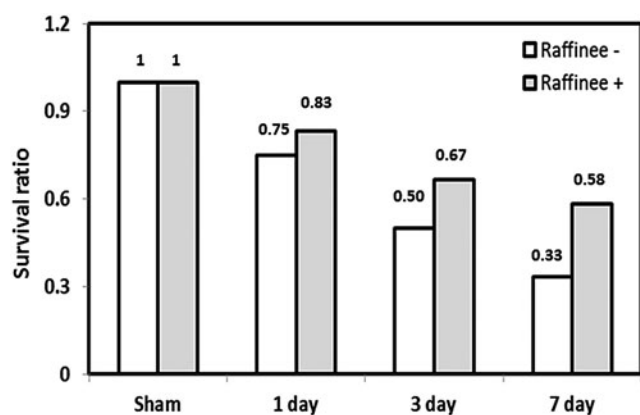


FIG. 6. Effects of treatment with Raffinee on survival of mice after pMCAO. Survival ratio of mice after pMCAO for 1, 3, or 7 days ($N=12$ per group). Value above bar is survival ratio mice number divided by 12.

with PBS (groups C, E, and G) (Fig. 5B–D). The pMCAO mice that had been treated with Raffinee for 1, 3, or 7 days had greatly higher survival ratios than those administrated with PBS for 1, 3, or 7 days, respectively (Fig. 6).

DISCUSSION

Stroke patients often suffered from long-term disability and enormous emotional and socioeconomic burden.^{36–38} Clinical evidence has indicated that 87% of strokes are ischemic.^{39,40} Owing to the very limited time window for effective intervention, less than 2% of ischemic stroke patients have the opportunity to receive thrombolytic therapy,¹⁷ motivating the development of neuroprotectants for ischemic stroke therapy. Notably, this study indicates the beneficial effects of Raffinee, a natural fermented product with strong antioxidant and neuroprotective activities, on PC12 cells by reducing hypoxia-induced apoptosis and on pMCAO mice by increasing levels of neurotrophic factors and reducing serum levels of inflammatory factors and brain infarct volume, greatly increasing survival rate, suggesting the neuroprotective potential of Raffinee in cases of acute ischemic stroke.

Ischemic stroke causes serious inflammatory and neurodegenerative injuries.⁴⁰ Several neurotrophic factors and inflammatory molecules are related to the etiology of ischemic stroke.^{13,14,41,42} NSE and S100B are known biomarkers for central nervous system (CNS) lesions.⁴³ Numerous inflammatory associated molecules, including MMPs, TNF- α , IL-6, and IL-10, are also related to ischemic stroke.^{7,9,11,44,45} NSE and S100 β , the so-called structural proteins of the CNS, are well-known biomarkers of CNS injury and are frequently used for prognosis in acute ischemic stroke.^{43,46,47} MMPs are important in ischemic stroke that participates in the injury and recovery processes, including tissue disruption, tissue formation, neuronal network remodeling, and blood–brain barrier integrity.^{10,11} MMP-9 is a crucial member of MMPs family. Elevated MMP-9 level and activity are strongly associated with bigger infarct volumes, greater stroke severity, and poor functional outcomes.⁴⁸ Various studies have also

indicated that TNF- α and IL-6 are pleiotropic cytokines that involve in both physiological and pathological processes, especially inflammation.⁴⁹ Elevated serum TNF- α and IL-6 levels are strongly related to acute cerebral infarction and disease severity.⁹ Indeed, various studies have found that both TNF- α and IL-6 play crucial roles in etiology of ischemic stroke and that serum TNF- α and IL-6 levels are reliable biomarkers in the prognosis of ischemic stroke.^{50,51} Interleukin 10 is known as an anti-inflammatory cytokine.⁷ IL-10 can reduce the activation and effector functions of various immune cells, such as T cells and macrophages, that inhibit inflammation.⁸ Moreover, IL-10 increases along with CNS pathology that promotes neuronal and glial cell survival.⁷ Accordingly, this study indicates that the administration of Raffinee increases serum S100 β and IL-10 levels and reduces the MMP-9 activity, serum TNF- α and IL-6 levels, and brain infarct volume in pMCAO mice. These findings demonstrated a beneficial effect of Raffinee in ischemic stroke.

Alternative medicines, such as herbal extracts, massage, dietary supplements, and acupuncture, has long been used for stroke prevention and recovery.⁵² Herbal medicines have been reported to be invaluable sources of new treatments for various human illness.⁵³ Many studies have reported the beneficial effects of herbal extracts or their active ingredients on different types of stroke by performing anti-inflammatory and neuroprotective effects in animal models with pMCAO.^{54–57} Similarly, the current study reveals that Raffinee significantly attenuated the inflammation and brain infarcted volume in pMCAO mice. Raffinee, cooperatively produced product by Ziel Enterprise and Toyo Hakko, is a natural fermented metabolite that is derived from a mixture of common crops.²⁶ The major ingredients of Raffinee are GMT that contains various kinds of polyphenols.²⁶ It exhibits powerful free radical scavenging activities.^{23,24} Notably, Raffinee exhibited antiatherogenic effects in a rabbit model and neuroprotective effects in an open-labeled clinical trial by improving motor and sensory function in SCI patients,^{23,24} suggesting superior antioxidant and neuroprotective activities of Raffinee. This study reveals its promising neuroprotective effects against permanent focal ischemia cerebral injury in a pMCAO experimental mice model and its potential as an alternative option in therapy for ischemic stroke. Hence, further research is needed to identify the bioactive ingredients of Raffinee and to obtain scientifically conclusive data of the effectiveness and safety of Raffinee in ischemic stroke patients.

Our earlier investigations have indicated the dominant antioxidant activity of Raffinee and demonstrated its neuroprotective properties on SCI, suggesting Raffinee as an effective adjuvant therapy to promote functional recovery from CNS injury.^{23,24} The current study further reveals the advantageous effects of Raffinee on a pMCAO mice model by reducing various inflammatory indicators and brain infarcted volume, suggesting that Raffinee could be a safe and effective alternative treatment for ischemic stroke. Additionally, a randomized double-blind placebo control study is strongly recommended for demonstrating the utility of Raffinee on pMCAO therapy in the future.

AUTHOR DISCLOSURE STATEMENT

No competing financial interests exist.

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REFERENCES

- WHO: *The World Health Report 2018*. WHO, Geneva, 2018.
- Feigin VL, Lawes CM, Bennett DA, Anderson CS: Stroke epidemiology: A review of population-based studies of incidence, prevalence, and case-fatality in the late 20th century. *Lancet Neurol* 2003;2:43–53.
- Mozzafarian D, Benjamin EJ, Go AS, *et al.*: American heart association statistics committee; stroke statistics subcommittee. Heart disease and stroke statistics-2016 update: A report from the American heart association. *Circulation* 2016;133:e38–e360.
- Hong KS, Saver JL: Quantifying the value of stroke disability outcomes: WHO global burden of disease project disability weights for each level of the modified rankin scale. *Stroke* 2009;40:3828–3833.
- Chan PH: Mitochondria and neuronal death/survival signaling pathways in cerebral ischemia. *Neurochem Res* 2004;29:1943–1949.
- Yang Y, Rosenberg GA: Blood-brain barrier breakdown in acute and chronic cerebrovascular disease. *Stroke* 2011;42:3323–3328.
- Strle K, Zhou JH, Shen WH, *et al.*: Interleukin-10 in the brain. *Crit Rev Immunol* 2001;21:427–449.
- Moore KW, de Waal Malefyt R, Coffman RL, O'Garra A: Interleukin-10 and the interleukin-10 receptor. *Annu Rev Immunol* 2001;19:683–765.
- Lin JZ, Miao KQ, Zhang HX, *et al.*: Change of early serum TNF- α and IL-6 levels in acute cerebral infarction and its significances. *Zhejiang Da Xue Xue Bao Yi Xue Ban* 2010;39:415–418.
- Yang Y, Rosenberg GA: Matrix metalloproteinases as therapeutic targets for stroke. *Brain Res* 2015;1623:30–38.
- Rempe RG, Hartz AMS, Bauer B: Matrix metalloproteinases in the brain and blood-brain barrier: Versatile breakers and makers. *J Cereb Blood Flow Metab* 2016;36:1481–1507.
- Bustamante A, Vilar-Bergua A, Guettier S, *et al.*: C-reactive protein in the detection of post-stroke infections: Systematic review and individual participant data analysis. *J Neurochem* 2017;141:305–314.
- Ahmad O, Wardlaw J, Whiteley WN: Correlation of levels of neuronal and glial markers with radiological measures of infarct volume in ischaemic stroke: A systematic review. *Cerebrovasc Dis* 2012;33:47–54.
- Glushakova OY, Glushakov AV, Miller ER, Valadka AB, Hayes RL: Biomarkers for acute diagnosis and management of stroke in neurointensive care units. *Brain Circ* 2016;2:28–47.
- Steiner T, Hacke W: Combination therapy with neuroprotectants and thrombolytics in acute ischaemic stroke. *Eur Neurol* 1998;40:1–8.
- Segura T, Calleja S, Jordan J: Recommendations and treatment strategies for the management of acute ischemic stroke. *Expert Opin Pharmacother* 2008;9:1071–1085.
- Goldstein LB, Rothwell PM: Advances in prevention and health services delivery 2007. *Stroke* 2008;39:258–260.
- Tang KS, Tan JS: The protective mechanisms of polydatin in cerebral ischemia. *Eur J Pharmacol* 2018;842:133–138.
- Shi L, Rocha M, Leak RK, *et al.*: A new era for stroke therapy: Integrating neurovascular protection with optimal reperfusion. *J Cereb Blood Flow Metab* 2018;38:2073–2091.
- Sze FK, Yeung FF, Wong E, Lau J: Does Danshen improve disability after acute ischaemic stroke? *Acta Neurol Scand* 2005;111:118–125.
- Yang XY, Wang LQ, Li JG, *et al.*: Chinese herbal medicine Dengzhan Shengmai capsule as adjunctive treatment for ischemic stroke: A systematic review and meta-analysis of randomized clinical trials. *Complement Ther Med* 2018;36:82–89.
- Liu S, Wu JR, Zhang D, *et al.*: Comparative efficacy of Chinese herbal injections for treating acute cerebral infarction: A network meta-analysis of randomized controlled trials. *BMC Complement Altern Med* 2018;18:120.
- Chen HY, Lin JM, Lin CC: Raffinee, a free radical scavenger, in the treatment of subacute stage brain and spinal cord lesions: A case report. *Am J Chin Med* 1998;26:97–108.
- Chen HY, Lin JM, Chuang HY, Chiu WT: Raffinee in the treatment of spinal cord injury: An open-labeled clinical trial. *Ann N Y Acad Sci* 2005;1042:396–402.
- Westerink RH, Ewing AG: The PC12 cell as model for neurosecretion. *Acta Physiol (Oxf)* 2008;192:273–285.
- Naito M, Wu X, Lin JM, *et al.*: Anti-atherogenic effects of fermented fresh coffee bean, soybean and rice bran extracts. *Food Sci Technol Res* 2003;9:170–175.
- Colak G, Filiano AJ, Johnson GV: The application of permanent middle cerebral artery ligation in the mouse. *J Vis Exp* 2011;53:3039.
- Shanbhag NC, Henning RH, Schilling L: Long-term survival in permanent middle cerebral artery occlusion: A model of malignant stroke in rats. *Sci Rep* 2016;6:28401.
- Gregersen R, Lambertsen KL, Finsen B: Microglia and macrophages are the major source of tumor necrosis factor in permanent middle cerebral artery occlusion in mice. *J Cereb Blood Flow Metab* 2000;20:53–65.
- Kuraoka M, Furuta T, Matsuwaki T, *et al.*: Direct experimental occlusion of the distal middle cerebral artery induces high reproducibility of brain ischemia in mice. *Exp Anim* 2009;58:19–29.
- US FDA Guidance for Industry: Estimating the Maximum Safe Starting Dose in Initial Clinical Trials for Therapeutics in Adult Healthy Volunteers. 2005. www.fda.gov/cder/guidance/index.htm. (Accessed September 1, 2018).
- Xiong Y, Zhu WZ, Zhang Q, Wang W: Observation of post-MCAO cortical inflammatory edema in rats by 7.0 Tesla MRI. *J Huazhong Univ Sci Technolog Med Sci* 2014;34:120–124.
- Cheng Q, Zhang Z, Zhang S, *et al.*: Human umbilical cord mesenchymal stem cells protect against ischemic brain injury in mouse by regulating peripheral immunoinflammation. *Brain Res* 2015;1594:293–304.
- Swanson RA, Morton MT, Tsao-Wu G, *et al.*: A semiautomated method for measuring brain infarct volume. *J Cereb Blood Flow Metab* 1990;10:290–293.
- Hsu TC, Chiu CC, Chang SC, *et al.*: Human parvovirus B19 VP1u protein as inflammatory mediators induces liver injury in naïve mice. *Virulence* 2016;7:110–118.
- Murray CJL, Lopez AD: Mortality by cause for eight regions of the world: Global burden of disease study. *Lancet* 1997;349:1269–1276.

37. Sarti C, Rastenyte D, Cepaitis Z, Tuomilehto J: International trends in mortality from stroke, 1968 to 1994. *Stroke* 2000;31:1588–1501.
38. Caro JJ, Huybrechts KF, Duchesne I: Management patterns and costs of acute ischemic stroke: An international study. *Stroke* 2000;31:582–590.
39. Hinkle JL, Guanci MM: Acute ischemic stroke review. *J Neurosci Nurs* 2007;39:285–293, 310.
40. Benjamin EJ, Blaha MJ, Chiuve SE, *et al.*: American heart association statistics committee and stroke statistics subcommittee. Heart disease and stroke statistics-2017 update: A report from the American heart association. *Circulation* 2017;135:e146–e603.
41. Lakhan SE, Kirchgessner A, Hofer M: Inflammatory mechanisms in ischemic stroke: Therapeutic approaches. *J Transl Med* 2009;7:97.
42. Jin R, Yang G, Li G: Inflammatory mechanisms in ischemic stroke: Role of inflammatory cells. *J Leukoc Biol* 2010;87:779–789.
43. Bloomfield SM, McKinney J, Smith L, Brisman J: Reliability of S100B in predicting severity of central nervous system injury. *Neurocrit Care* 2007;6:121–138.
44. Tuttolomondo A, Di Sciacca R, Di Raimondo D, *et al.*: Inflammation as a therapeutic target in acute ischemic stroke treatment. *Curr Top Med Chem* 2009;9:1240–1260.
45. Mizuma A, Yenari MA: Anti-inflammatory targets for the treatment of reperfusion injury in stroke. *Front Neurol* 2017;8:467.
46. Barone FC, Clark RK, Price WJ, *et al.*: Neuron-specific enolase increases in cerebral and systemic circulation following focal ischemia. *Brain Res* 1993;623:77–82.
47. Missler U, Wiesmann M, Friedrich C, Kaps M: S-100 protein and neuron-specific enolase concentrations in blood as indicators of infarction volume and prognosis in acute ischemic stroke. *Stroke* 1997;28:1956–1960.
48. Ramos-Fernandez M, Bellolio MF, Stead LG: Matrix metalloproteinase-9 as a marker for acute ischemic stroke: A systematic review. *J Stroke Cerebrovasc Dis* 2011;20:47–54.
49. Zhang JM, An J: Cytokines, inflammation, and pain. *Int Anesthesiol Clin* 2007;45:27–37.
50. Shaafi S, Sharifipour E, Rahmanifar R, *et al.*: Interleukin-6, a reliable prognostic factor for ischemic stroke. *Iran J Neurol* 2014;13:70–76.
51. Klimiec E, Kowalska K, Pasinska P, *et al.*: Reduced release of TNF α and IP-10 after ex vivo blood stimulation with endotoxin is associated with poor outcome after stroke. *Cytokine* 2018;102:51–54.
52. Yeh ML, Chiu WL, Wang YJ, Lo C: An investigation of the use of traditional Chinese medicine and complementary and alternative medicine in stroke patients. *Holist Nurs Pract* 2017;31:400–407.
53. Gaire BP: Herbal medicine in ischemic stroke: Challenges and prospective. *Chin J Integr Med* 2018;24:243–246.
54. Chan SJ, Wong WS, Wong PT, Bian JS: Neuroprotective effects of andrographolide in a rat model of permanent cerebral ischemia. *Br J Pharmacol* 2010;161:668–679.
55. Liang W, Huang X, Chen W: The effects of Baicalin and Baicalein on cerebral ischemia: A review. *Aging Dis* 2017;8:850–867.
56. Sun R, Song Y, Li S, *et al.*: Levo-tetrahydropalmatine attenuates neuron apoptosis induced by cerebral ischemia-reperfusion injury: Involvement of c-Abl activation. *J Mol Neurosci* 2018;65:391–399.
57. Zhao C, Zhang C, Xing Z, *et al.*: Pharmacological effects of natural Ganoderma and its extracts on neurological diseases: A comprehensive review. *Int J Biol Macromol* 2019;121:1160–1178.